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Modul : Supervised Learning

Prediction Measurement

**Inteligensi Buatan
(Artificial Intelligence)**



Prediction Measurement

Supervised
Learning



“Correct Class”

Prediction based
on Model/
Hypothesis from
Learning

		Prediction	
		True	False
Reality	True	Tp True-positive	Fn False-negative
	False	Fp False-positive	Tn True-negative



Example

Instance	Correct Class	Prediction	
1	+	+	TP
2	-	-	TN
3	-	+	FP
4	+	-	FN
5	-	+	FP
6	+	-	FN
7	+	+	TP
8	-	-	TN
9	-	-	TN
10	+	+	TP



Accuracy

		Prediction	
		True	False
Reality	True	Tp True-positive	Fn False-negative
	False	Fp False-positive	Tn True-negative

Fraction of all correct predictions over all predicted instances

$$Accuracy = \frac{Tp + Tn}{Tp + Fp + Tn + Fn}$$



Precision

		Prediction	
		True	False
Reality	True	Tp True-positive	Fn False-negative
	False	Fp False-positive	Tn True-negative

Fraction of positive predictions that are correct

$$Precision = \frac{Tp}{Tp + Fp}$$

Recall

		Prediction	
		True	False
Reality	True	Tp True-positive	Fn False-negative
	False	Fp False-positive	Tn True-negative

Fraction of positive instances that are correctly predicted
(retrieved/caught)

$$\text{Recall} = \frac{Tp}{Tp + Fn}$$



F1-Score

$$F_1 = \frac{2}{\text{recall}^{-1} + \text{precision}^{-1}} = 2 \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$$

Harmonic mean of precision and recall

Exercise: Find accuracy, precision and recall

Instance	Correct Class	Prediction
1	+	+
2	-	-
3	-	+
4	+	-
5	-	+
6	+	-
7	+	+
8	-	-
9	-	-
10	+	+
11	+	-
12	-	-
13	-	+
14	+	-



MultiClass Problem

No	Actual	Predicted	Match
1	Airplane	Airplane	✓
2	Car	Boat	✗
3	Car	Car	✓
4	Car	Car	✓
5	Car	Boat	✗
6	Airplane	Boat	✗
7	Boat	Boat	✓
8	Car	Airplane	✗
9	Airplane	Airplane	✓
10	Car	Car	✓

		Predicted		
		 Airplane	 Boat	 Car
Actual	 Airplane	2	1	0
	 Boat	0	1	0
	 Car	1	2	3

Source: <https://iamirmasoud.com/2022/06/19/understanding-micro-macro-and-weighted-averages-for-scikit-learn-metrics-in-multi-class-classification-with-example/>

Label	True Positive (TP)	False Positive (FP)	False Negative (FN)	Precision	Recall	F1 Score
 Airplane	2	1	1	0.67	0.67	$2 * (0.67 * 0.67) / (0.67 + 0.67)$ = 0.67
 Boat	1	3	0	0.25	1.00	$2 * (0.25 * 1.00) / (0.25 + 1.00)$ = 0.40
 Car	3	0	3	1.00	0.50	$2 * (1.00 * 0.50) / (1.00 + 0.50)$ = 0.67



MultiClass Problem

No	Actual	Predicted	Match
1	Airplane	Airplane	✓
2	Car	Boat	✗
3	Car	Car	✓
4	Car	Car	✓
5	Car	Boat	✗
6	Airplane	Boat	✗
7	Boat	Boat	✓
8	Car	Airplane	✗
9	Airplane	Airplane	✓
10	Car	Car	✓

micro-F1 = micro-precision = micro-recall = accuracy

Label	Per-Class F1 Score	Macro-Averaged F1 Score
 Airplane	0.67	$\frac{0.67 + 0.40 + 0.67}{3}$ = 0.58
 Boat	0.40	
 Car	0.67	

Label	Per-Class F1 Score	Support	Support Proportion	Weighted Average F1 Score
 Airplane	0.67	3	0.3	$(0.67 * 0.3) + (0.40 * 0.1) + (0.67 * 0.6)$ = 0.64
 Boat	0.40	1	0.1	
 Car	0.67	6	0.6	
Total	-	10	1.0	

Label	True Positive (TP)	False Positive (FP)	False Negative (FN)	Micro-Averaged Values
 Airplane	2	1	1	Precision = $\frac{6}{6+4} = \mathbf{0.60}$ Recall = $\frac{6}{6+4} = \mathbf{0.60}$ F1 Score = $\frac{6}{6 + \frac{1}{2}(4+4)} = \mathbf{0.60}$
 Boat	1	3	0	
 Car	3	0	3	
TOTAL	6	4	4	



THANK YOU

